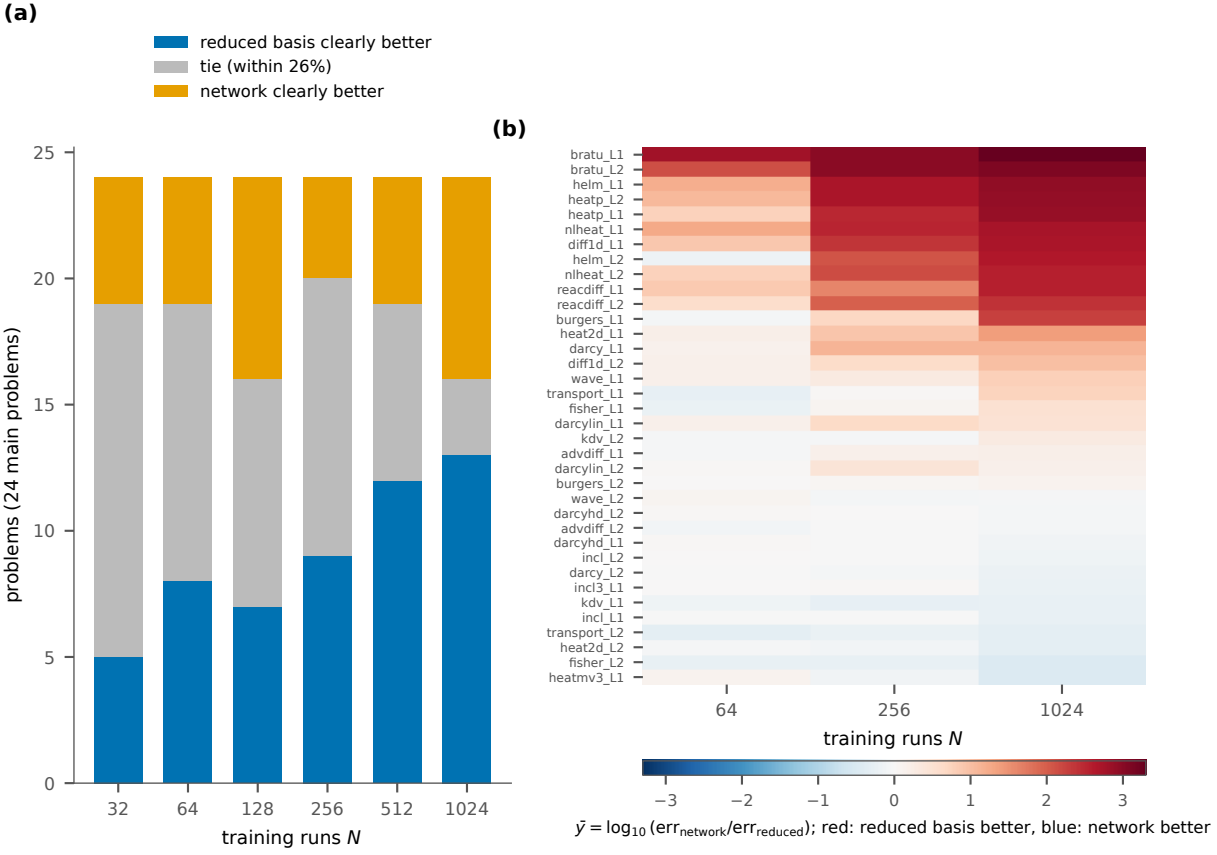
(a)
reduced basis clearly better
tie (within 26%)
network clearly better
problems (24 main problems)
0
5
10
15
20
25
32
64
128
256
512
1024
training runs N
(b)
bratu_L1
bratu_L2
helm_L1
heatp_L2
heatp_L1
nlheat_L1
diff1d_L1
helm_L2
nlheat_L2
reacdiff_L1
reacdiff_L2
burgers_L1
heat2d_L1
darcy_L1
diff1d_L2
wave_L1
transport_L1
fisher_L1
darcylin_L1
kdv_L2
advdiff_L1
darcylin_L2
burgers_L2
wave_L2
darcyhd_L2
advdiff_L2
darcyhd_L1
incl_L2
darcy_L2
incl3_L1
kdv_L1
incl_L1
transport_L2
heat2d_L2
fisher_L2
heatmv3_L1
64
256
1024
training runs N
−3
−2
−1
0
1
2
3
$\bar{y} = \log_{10}(\mathrm{err}_{\mathrm{network}}/\mathrm{err}_{\mathrm{reduced}})$; red: reduced basis better, blue: network better